EE301 HW4 Solutions

Problem 1:

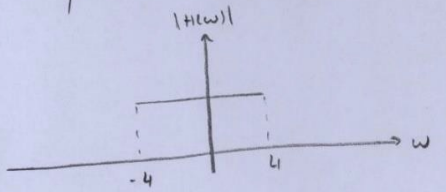
First find F.T. of $h(t)$.

We know that $F\left\{ \text{rect}\left(\frac{t}{T}\right)\right\} = \frac{\sin\left(\frac{T}{2}\omega\right)}{\omega}$ by duality,

$$F\left\{ \frac{2\sin\left(\frac{T}{2}t\right)}{t} \right\} = 2\pi\,\text{rect}\left(-\frac{\omega}{T}\right), \text{ therefore}$$

$$F\left\{ \frac{1}{\pi t}\sin(4t) \right\} = \text{rect}\left(\frac{\omega}{8}\right), \text{ then use time-shift property}$$

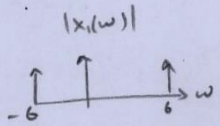
$$F\left\{ \frac{1}{\pi(t-1)}\sin(4(t-1)) \right\} = e^{-j\omega}\,\text{rect}\left[\frac{\omega}{8}\right] = H(\omega)$$



a) $x_1(t) = \cos\left(6t + \frac{\pi}{2}\right)$

$$X_1(\omega) = \pi e^{j\frac{\pi}{2}}\left[\delta(\omega-6) + \delta(\omega+6)\right]$$

$$Y_1(\omega) = X_1(\omega)H(\omega) = 0$$



b) $x_2(t) = \frac{\sin(4(t+1))}{\pi(t+1)}$, similar to $H(\omega)$, $X_2(\omega)$ can be found as

$$X_2(\omega) = e^{j\omega}\,\text{rect}\left[\frac{\omega}{8}\right]$$

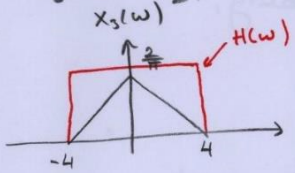
$$Y_2(\omega) = X_2(\omega)H(\omega) = e^{-j\omega}e^{j\omega}\,\text{rect}\left(\frac{\omega}{8}\right)\text{rect}\left(\frac{\omega}{8}\right) = \text{rect}\left(\frac{\omega}{8}\right)$$

$$y_2(t) = F^{-1}\left\{ \text{rect}\left(\frac{\omega}{8}\right)\right\} = \frac{\sin 4t}{\pi t}$$

c) $x_3(t) = \left(\frac{\sin 2t}{\pi t}\right)\left(\frac{\sin 2t}{\pi t}\right)$ let $\frac{\sin 2t}{\pi t} = z(t)$

$z(\omega) = \text{rect}\left(\frac{\omega}{4}\right)$

$X_3(\omega) = \frac{1}{2\pi} z(\omega) * z(\omega) = \frac{1}{2\pi} T(\omega)$ where $T(\omega) = \begin{cases} 4-\omega, & 0<\omega<4 \\ 4+\omega & -4<\omega<0 \\ 0 & \text{o.w} \end{cases}$



$Y_3(\omega) = H(\omega)X_3(\omega) = e^{-j\omega} X_3(\omega)$

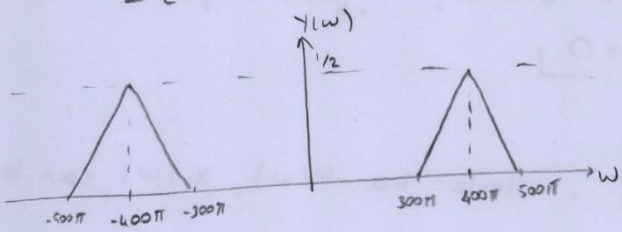
$y_3(t) = \left[\frac{\sin[2(t-1)]}{\pi(t-1)}\right]^2$

Q.2

a) $y(t) = x(t)\cos(400\pi t)$

$Y(\omega) = \frac{1}{2\pi} X(\omega) * [\pi\delta(\omega-400\pi) + \pi\delta(\omega+400\pi)]$

$Y(\omega) = \frac{1}{2}[X(\omega-400\pi) + X(\omega+400\pi)]$



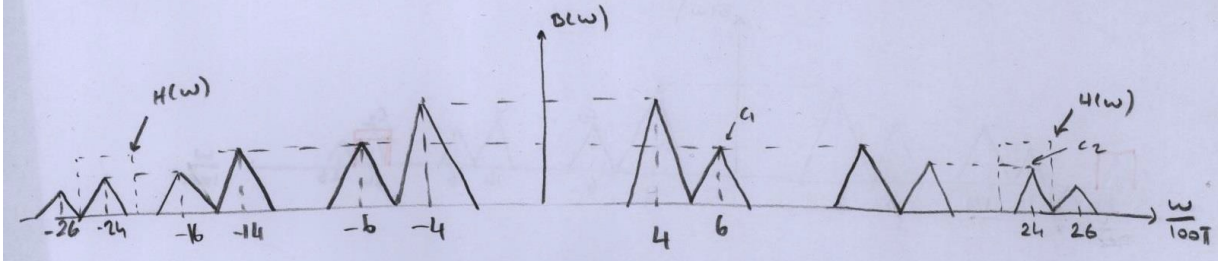
b) $b(t) = y(t)c(t)$, fundamental frequency of $c(t)$ is 1000π rad/s

we can express $c(t)$ as $c(t) = \sum c_k e^{jk1000\pi t}$

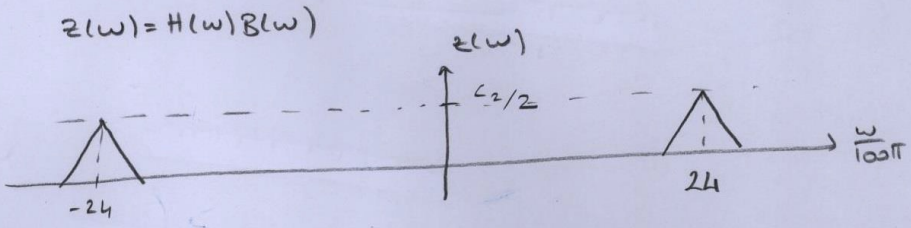
$F\{c(t)\} = C(\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k \delta(\omega - 1000\pi k)$

$B(\omega) = \frac{1}{2\pi} Y(\omega) * \left[2\pi \sum_k c_k \delta(\omega - k1000\pi)\right]$

$= \sum_{k=-\infty}^{\infty} c_k Y(\omega - k1000\pi)$



$Z(\omega) = H(\omega)B(\omega)$



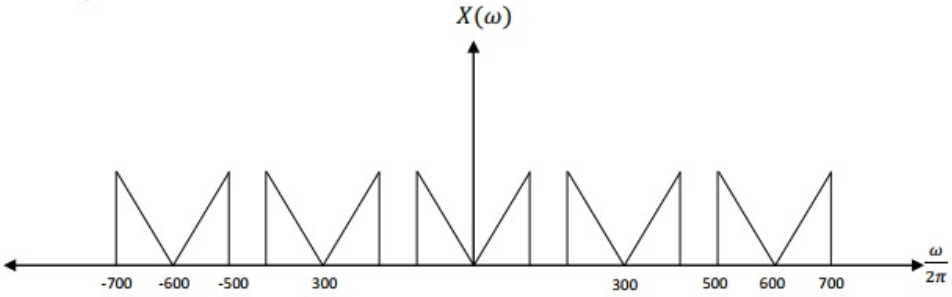
c) $Z(\omega) = \frac{c_2}{2} X(\omega - 2400\pi) + \frac{c_{-2}}{2} X(\omega + 2400\pi)$

$$z(t) = \frac{c_2}{2} x(t) e^{j2400\pi t} + \frac{c_{-2}}{2} x(t) e^{j2400\pi t}$$

Problem 3.

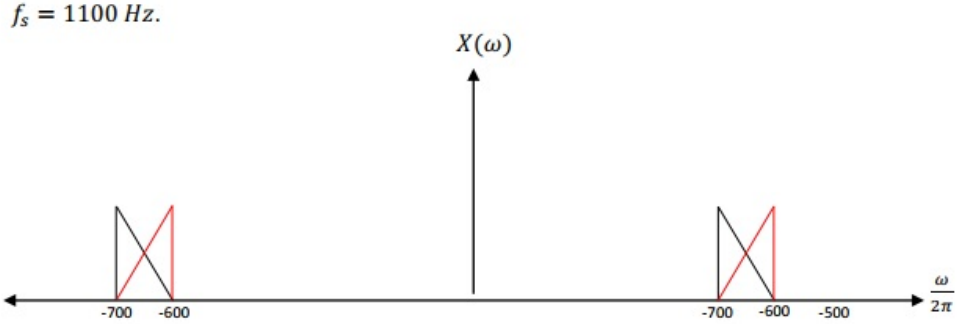
By the sampling theorem; $X_s(w) = f_s \sum_{k=\infty}^{-\infty} X(w - 2\pi f_s k)$

$f_s = 300\ Hz$



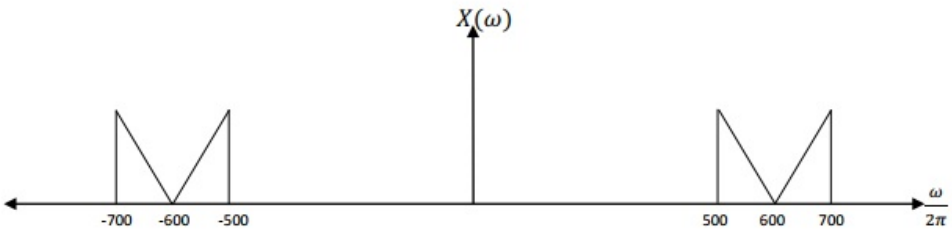
Since there is no overlap, 300 Hz can be used as sampling frequency.

$f_s = 1100\ Hz.$



Since there is an overlap, 1100 Hz cannot be used for sampling.

$f_s = 1200\ Hz.$

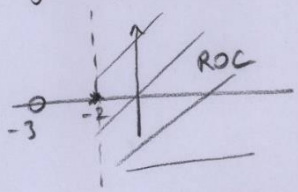


Since there is no overlap, 1200 Hz can be used as sampling frequency.

Q.4

a) $H(s) = \dfrac{3+s}{(2+s)^2}$ $\quad z = -3 \quad p_1 = -2 \quad p_2 = -2$

Since the system is stable ROC should include $j\omega$-axis.



since ROC extends rightward from the rightmost pole, then the system is causal

$$H(s) = \cancel{\frac{2+s}{(2+s)^2}} + \frac{1}{(2+s)^2} = \frac{1}{2+s} + \frac{1}{(2+s)^2}$$

$$h(t) = \left(e^{-2t} + te^{-2t}\right) u(t)$$

b) $X(s) = 1 + 2\dfrac{1}{s+1} + \dfrac{1}{(s+1)^2} = \dfrac{(s+1)^2 + 2s + 2 + 1}{(s+1)^2} = \dfrac{(s+2)^2}{(s+1)^2}$

$$Y(s) = X(s)H(s) = \frac{(s+2)^2}{(s+1)^2} \cdot \frac{3+s}{(s+2)^2} = \frac{3+s}{(s+1)^2}$$

$$Y(s) = \frac{1}{(s+1)} + \frac{2}{(s+1)^2}$$

$$y(t) = e^{-t} + 2te^{-t}$$

Q.5

$$X(s) = \frac{s^2}{(s+1)(s^2+4s+5)} = \frac{s^2}{(s+1)\left[(s^2+4s+4)+1\right]} = \frac{s^2}{(s+1)\left[(s+2)^2+1\right]}$$

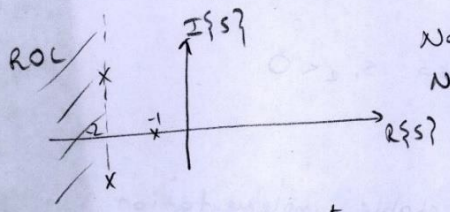
$$X(s) = \frac{s^2}{(s+1)(s+2-j)(s+2+j)} = \frac{A}{(s+1)} + \frac{B}{(s+2-j)} + \frac{B^*}{(s+2+j)}$$

$$A = (s+1)X(s)\Big|_{s=-1} = \frac{s^2}{(s+2-j)(s+2+j)}\Big|_{s=-1} = \frac{1}{2}$$

$$B = (s+2-j)X(s)\Big|_{s=-2+j} = \frac{s^2}{(s+1)(s+2+j)}\Big|_{s=-2+j} = \frac{3+j4}{(j-1)\,j2} = \frac{1}{4} + j\frac{7}{4}$$

$$X(s) = \frac{1}{2}\frac{1}{(s+1)} + B\frac{1}{(s+2-j)} + B^*\frac{1}{(s+2+j)}$$

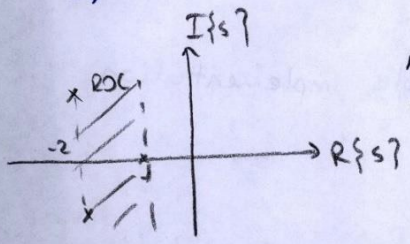
a) $R_x : R\{s\} < -2$



Non-causal
Not-stable

$$x(t) = -\frac{1}{2}e^{-t}u(-t) - B\,e^{-(2-j)t}u(-t) - B^*\,e^{-(2+j)t}u(-t)$$

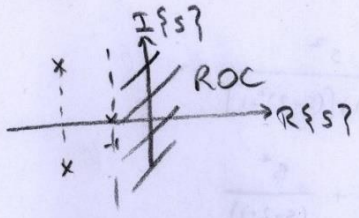
b) $R_x : -2 < R\{s\} < -1$



Non-causal
Not-stable

$$x(t) = -\frac{1}{2}e^{-t}u(-t) + Be^{-(2-j)t}u(t) + B^*e^{-(2+j)t}u(t)$$

c) $R_x : -1 < R\{s\}$



causal and stable

$$x(t) = \frac{1}{2} e^{-t} u(t) + B e^{-(2-j)t} u(t) + B^* e^{-(2+j)t} u(t)$$

Q.6

write the differential equation in s-domain

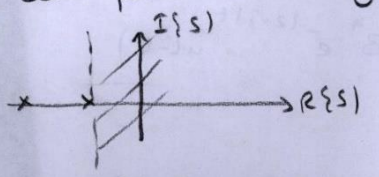
$$s^2 Y(s) + \frac{R}{L} s Y(s) + \frac{1}{LC} Y(s) = \frac{1}{LC} X(s)$$

$$\left(s^2 + \frac{R}{L} s + \frac{1}{LC}\right) Y(s) = \frac{1}{LC} X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{1/LC}{s^2 + \frac{R}{L} s + \frac{1}{LC}}$$ the poles are $$s_{1,2} = \frac{-\frac{R}{L} \mp \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}}{2}$$

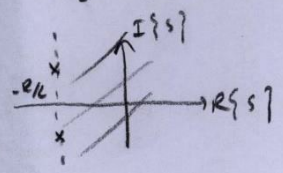
If $\frac{R^2}{L^2} - \frac{4}{LC} \geq 0$ $\quad \frac{R}{L} \geq \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}} \geq 0$, so $s_{1,2} < 0$

both poles have negative values



causal and stable implementation exists.

If $\frac{R^2}{L^2} - \frac{4}{LC} < 0$ then $s_{1,2} = \frac{-\frac{R}{L} \mp j\sqrt{\frac{4}{LC} - \frac{R^2}{L^2}}}{2}$ $\quad Re\{s_{1,2}\} = -\frac{R}{L} < 0$



causal and stable implementation exists.

$\therefore$ For all cases causal and stable implementation is possible.

Q.7

a) $y[n] - \frac{5}{2} y[n-1] + y[n-2] = x[n]$

$H(z) - \frac{5}{2} z^{-1} H(z) + z^{-2} H(z) = 1$

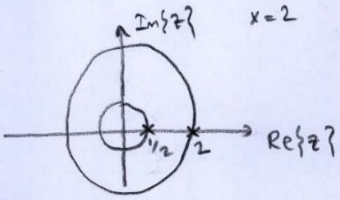
$$H(z) = \frac{1}{\left(1 - \frac{5}{2} z^{-1} + z^{-2}\right)} = \frac{2z^2}{2z^2 - 5z + 2} = \frac{2z^2}{(2z-1)(z-2)} = \frac{2}{(2-z^{-1})(1-2z^{-1})}$$

$$H(z) = \frac{1}{(1-\frac{1}{2}z^{-1})} \cdot \frac{1}{(1-2z^{-1})} = \frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{(1-2z^{-1})}$$

$$A = (1-\tfrac{1}{2}z^{-1}) H(z) \Big|_{z=1/2} = -\frac{1}{3}$$

$$B = (1-2z^{-1}) H(z) \Big|_{x=2} = \frac{4}{3}$$

$$H(z) = -\frac{1}{3} \cdot \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{4}{3} \cdot \frac{1}{1-2z^{-1}}$$



$$h(t) = -\frac{1}{3} (1/2)^n u[n] + \frac{4}{3} 2^n u[n]$$

b) $H(z) = \frac{Y(z)}{X(z)}$ $\quad X(z) = \frac{1}{(1-1/3 z^{-1})} - \frac{1}{(1-2z^{-1})} = \frac{1 - 2z^{-1} - 1 + \frac{1}{3}z^{-1}}{(1-\frac{1}{3}z^{-1})(1-2z^{-1})}$

$$Y(z) = 5 \frac{1}{(1-\frac{1}{3}z^{-1})} - 5 \frac{1}{(1-2z^{-1})} = 5 \frac{1 - \frac{2}{3}z^{-1} - 1 + \frac{1}{3}z^{-1}}{(1-\frac{1}{3}z^{-1})(1-\frac{2}{3}z^{-1})}$$

$$H(z) = 5 \frac{\frac{-1/3\, z^{-1}}{(1-1/3 z^{-1})(1-\frac{2}{3}z^{-1})}}{\frac{-5/3}{(1-\frac{1}{3}z^{-1})(1-2z^{-1})}} = \frac{1-2z^{-1}}{1-\frac{2}{3}z^{-1}} = \frac{Y(z)}{X(z)}$$

$$y(n) - \frac{2}{3} y[n-1] = x[n] - 2x(n-1)$$

Q.8

a) $\left(\frac{5}{6}\right)^n u[n-6] = \left(\frac{5}{6}\right)^6 \left(\frac{5}{6}\right)^{n-6} u[n-6]$

use time shift property

$$\mathcal{Z}\left\{\left(\frac{5}{6}\right)^6 \left(5/6\right)^{n-6} u[n-6]\right\} = \left(\frac{5}{6}\right)^6 \frac{z^{-6}}{1-\frac{5}{6}z^{-6}}$$

b) use differentiation in z-domain property

$$\mathcal{Z}\{n\,x(n)\} = -z\frac{d}{dz}X(z) = -z\frac{d}{dz}\left(\frac{1}{1-z^{-1}}\right) = -z\frac{-1}{(z-1)^2}$$

$$\mathcal{Z}\{n\,x(n)\} = \frac{z^{-1}}{(1-z^{-1})^2}$$

c) Let $x(n) = u[-n]$

$$\mathcal{Z}\{n\,u(-n)\} = \mathcal{Z}\{n\,x[n]\} = -z\frac{d}{dz}X(z)$$

$X(z) = \mathcal{Z}\{u(-n)\} = \frac{1}{1-z}$ by time shift property

$$\mathcal{Z}\{n\,u(-n)\} = -z\frac{d}{dz}\left(\frac{1}{1-z}\right) = -z\frac{1}{(1-z)^2} = -\frac{z}{(1-z)^2}$$

$$= -\frac{z^{-1}}{(1-z^{-1})^2}$$